

User Guide for Sumo Project application:

Welcome to the user guide for our Sumo Project application. This guide will explain how to install and use our created application with added pictures for clarity:

Overview:

1. First steps
2. Import own simulations
3. GUI
4. Export

1. First steps

Our application is a multithreaded, GUI based Java TraaS API wrapper, which starts up a Sumo simulation and connects to it, adding objects to the simulation in real time. You can also export the collected data with several styles and parameters. It is built on Java 25 and expected to work on later versions of Java.

Dependencies:

- Java 25 (or later)
- SUMO
- Traas.jar (Is installed with SUMO by default)

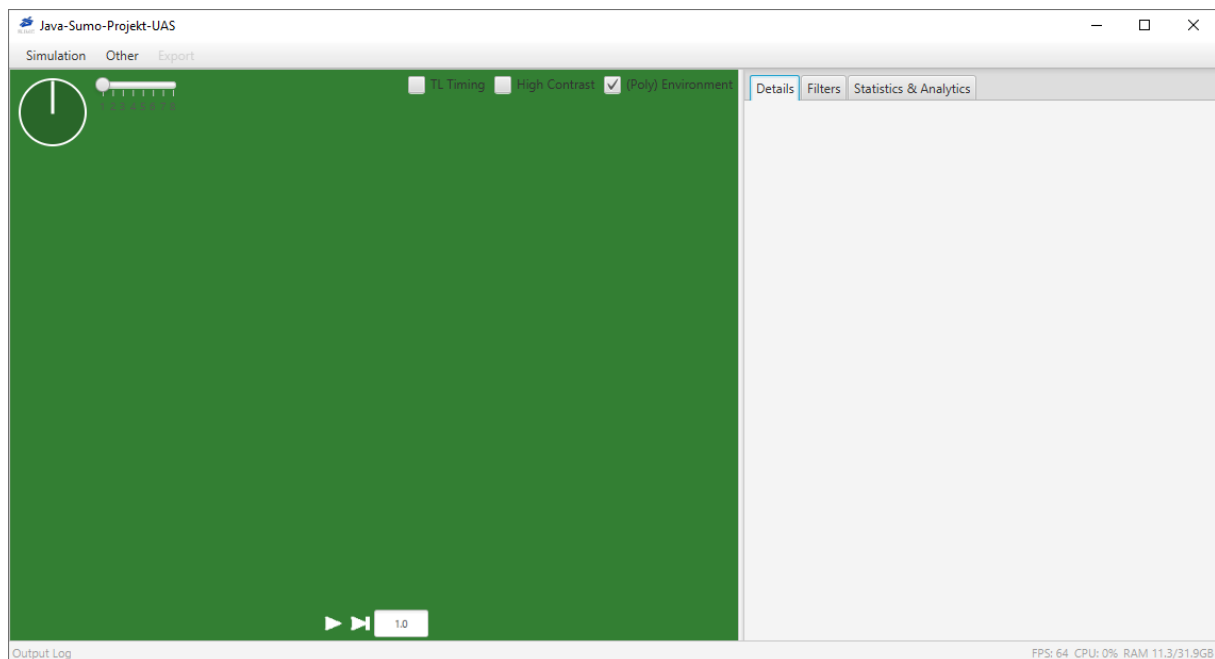
1.1 Installation:

1. Download and install Java from the official Java website (choose version 25 or above):
<https://www.java.com/de/download/manual.jsp>
2. Download and install SUMO from the official SUMO website:
<https://sumo.dlr.de/docs/Downloads.php>
3. Download our program from the releases tab from our GitHub:
(link is missing since we don't have a release yet)

You may also manually compile the program. The instructions on how to do that are on our GitHub project page, explanation will not be provided in this user guide.

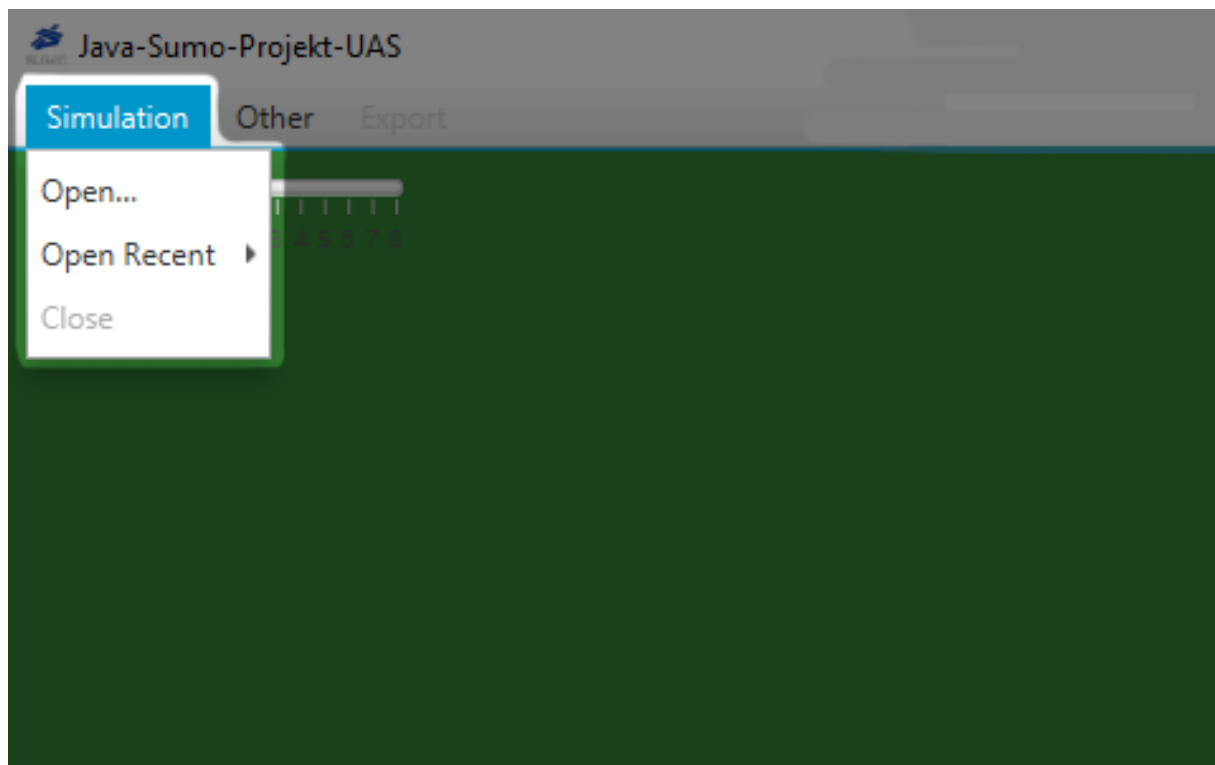
2. Import own simulations

Once all dependent programs have been installed, you are now ready to launch the application:



This is what you should see if the application has been launched correctly.

To select a desired simulation, navigate to the “Simulation” label on the top left side.

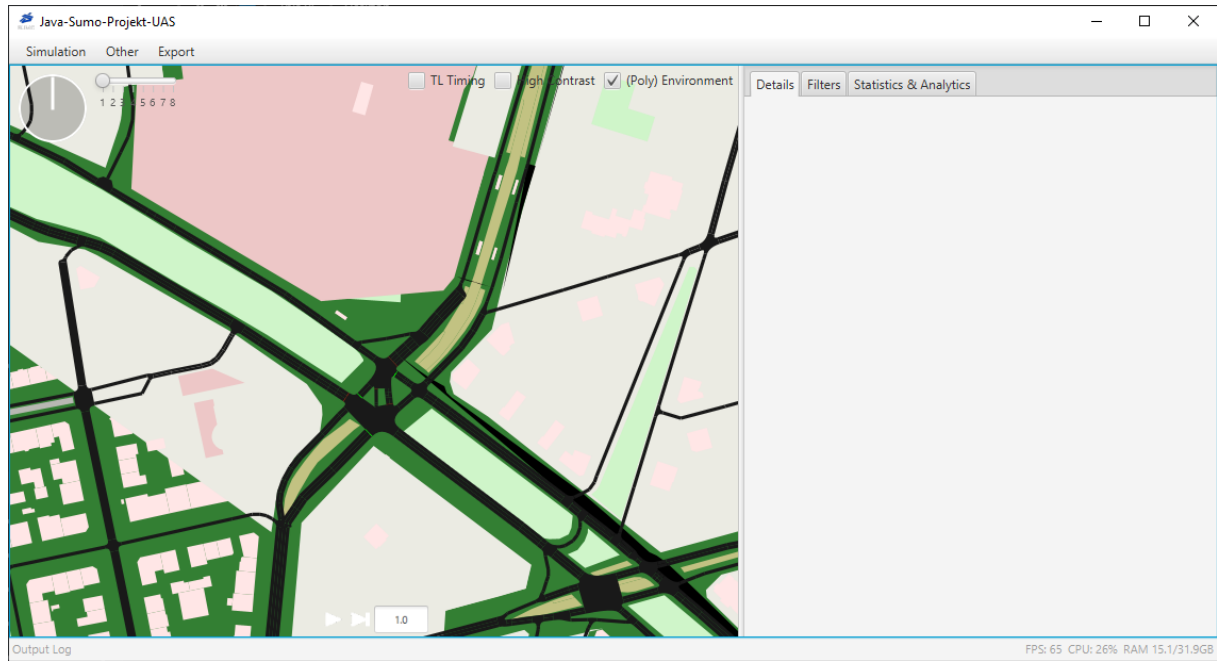


Once there, you will find an option called "Open..." with which you can select a file to open on your disk. You can only select files with the type ".sumocfg" or ".net.xml & .rou.xml". If you wish to select a network file, you must select a route file to go along with it; otherwise, the program will not work.

A “Open Recent” button is also available, allowing you to open the recently accessed simulation files.

If you wish to close the current simulation, use the “Close” button under “Simulation”

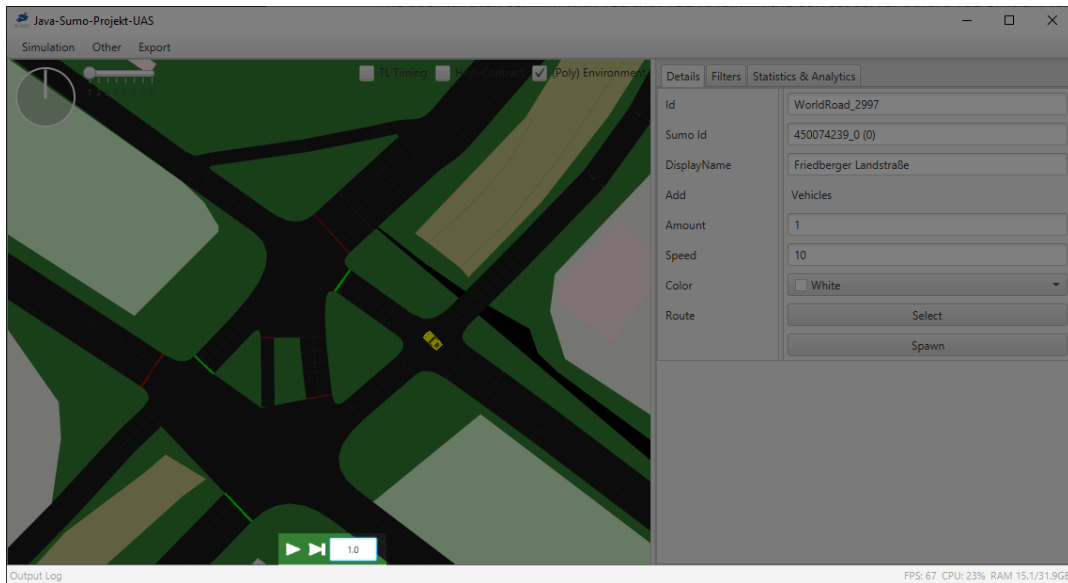
Once a valid file has been selected, the road network should appear in the world:



3. GUI

We aimed to make the GUI as intuitive and feature rich as possible. The following is a guide on how to use its features.

3.1 Simulation controls



On the bottom of the simulation view, you have 2 buttons and one input field:

 Is the play button to play the simulation. Once clicked, the button will display a pause symbol, allowing you to pause the simulation.

 Is the next-frame button used to skip one step forward in the simulation

 The input field is used to set the simulation speed (1.0 being 1 second). The highest number of seconds that is allowed is 25.

The “Output Log” text in the bottom left can be clicked, which will open a new debugging window, that shows you the performed actions.

On the bottom right, you can observe the current FPS, the system CPU load and the RAM that is currently allocated. (The CPU percentage may not always be accurate, depending on the operating system that the program is running on.)

To drag the simulation around, click and drag on the world with either your left or right mouse button.

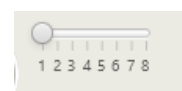
To zoom in and out, use your scroll wheel.

The size of the panels can be adjusted by dragging the edge in the desired direction.

To rotate, hold Alt + the left mouse button and drag in the direction you would like to rotate in; you will see this indicator in the top left moving in the direction that you rotate in.



To change the size of the vehicles, use the slider that is located next to the rotation indicator.



3.2 Object controls (Details tab)

When you click on a object like a car or a road, you get the information about the object displayed on the right side of the screen under the “Details” tab.

Objects that can be clicked on include:

- Roads
- Vehicles
- Traffic lights

Some of the fields contained in these menus are editable by the user. These fields will be marked **green**; fields that cannot be edited will be marked **red**. We recommend pausing the simulation when making any changes to the simulation. Changes in the vehicles tab can be applied by pressing enter.

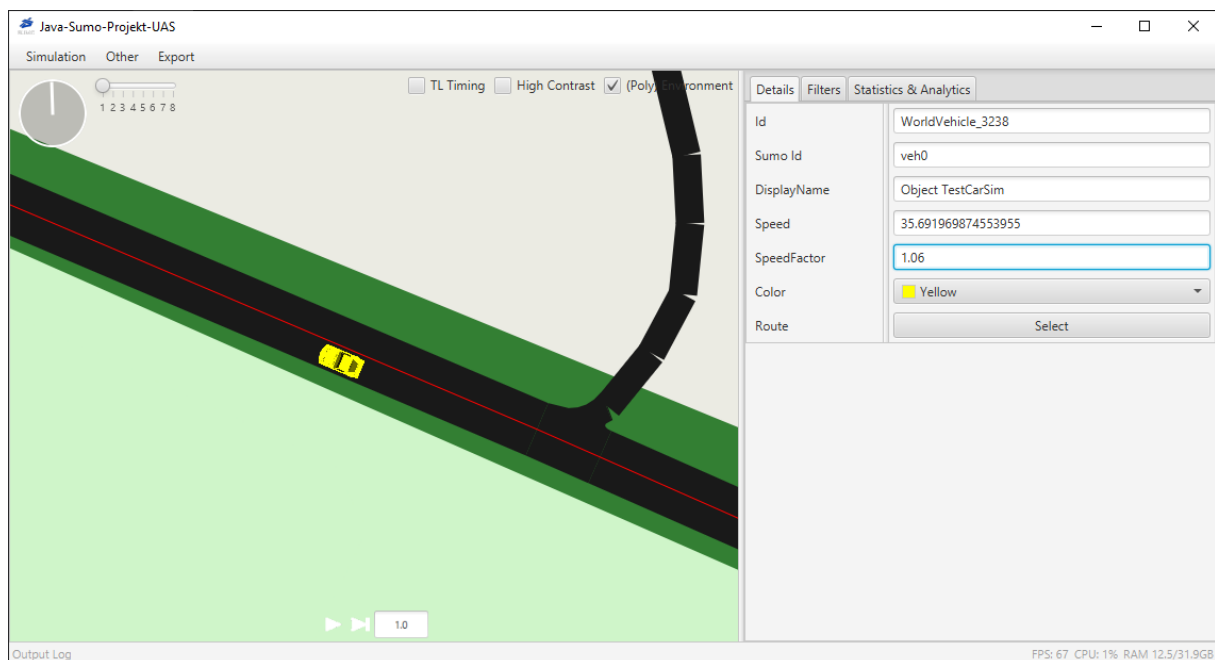
When having clicked on a car, the following information is displayed:

1. The ID that is assigned by our program
2. The SUMO ID
3. The display name
4. The speed
5. The Speed factor, which is the speed multiplier

The color of the vehicle can be changed with the color selector, which appears when you click on the color.

The route selection allows you to click onto any road to change the route of that vehicle to have it as its destination.

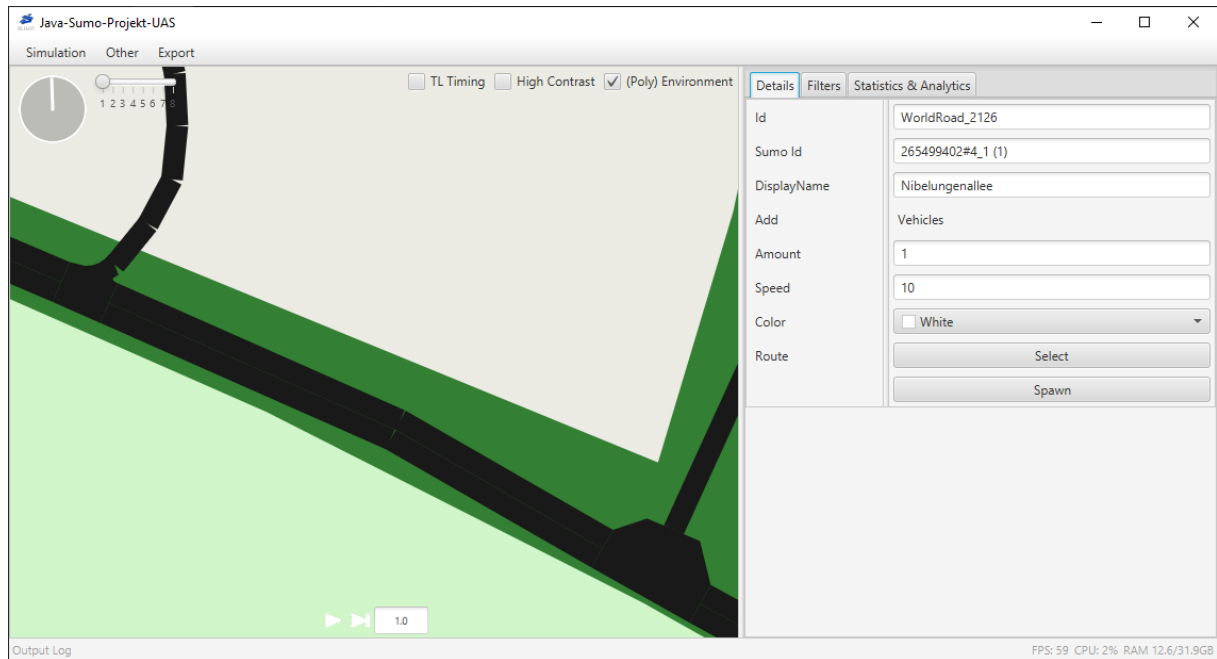
In the simulation window a red line will be displayed, which represents the route of that vehicle.



Having clicked on a road, the following information will be displayed:

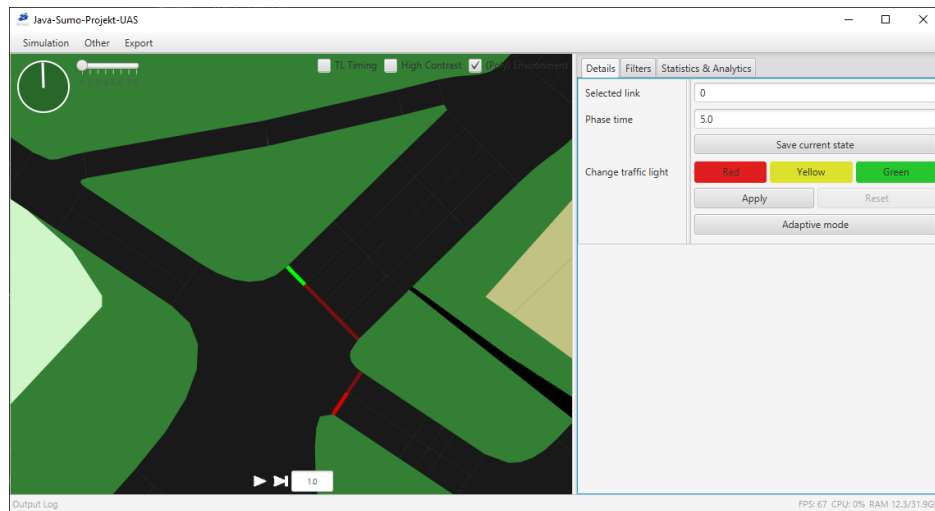
1. The ID that is assigned by our program
2. The SUMO ID
3. The display name (in this case, the real name of the street)

The ability to spawn vehicles on the selected road is available under these fields. You can specify the number of cars that will be spawned, the speed, the color and the route. After setting all these parameters and having clicked the spawn button, the cars will be spawned one after another if there is space to spawn them.



If you click on a traffic light, the following information will be displayed:

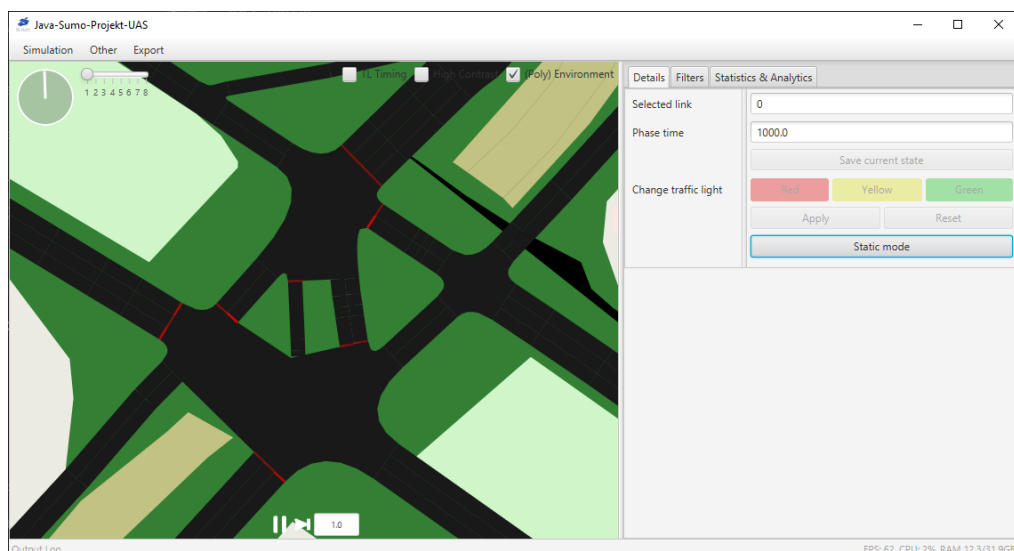
1. **The selected link** (which is the selected traffic light of that junction)
2. **The phase time** (which is how long it takes for the junction to switch to another phase)



Below you will find the traffic light controls:

The traffic light controls allow you to change the phase time and the color. Phase time is set for the entire junction, while color only changes the currently selected traffic light. You can also save and later reload to the saved state by using the “Save current state” and the “Reset” buttons. If no state for that junction is saved, the reset button will be grayed out. Once you have changed the traffic light to the desired color and/or changed the phase time, click the “Apply” button to apply the changes. The changes will occur on the next step of the simulation.

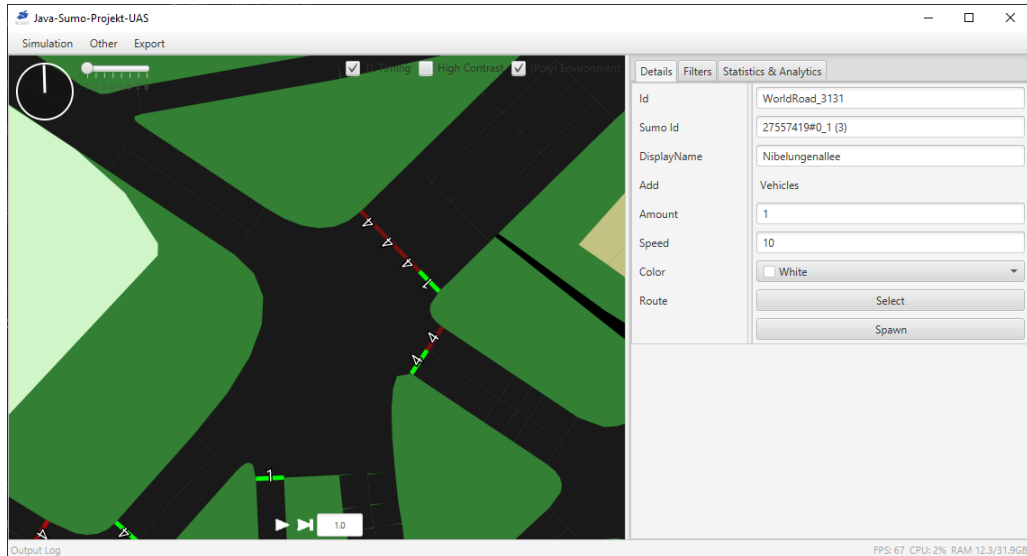
The “Adaptive mode” button activates an automatic detection for the vehicles. It will turn all traffic lights at a junction red and look for a halting vehicle at every connected traffic light, every 10 steps. If it finds any, the traffic light(s) with the most cars, will be set to green for 10 steps, after which, the junction is set to red again. The button automatically saves the state of the junction it had once the adaptive mode had been engaged. All other buttons will be grayed out. If you wish to go back to the static mode, click on the “Static mode” button and the simulation will return to the last saved state.



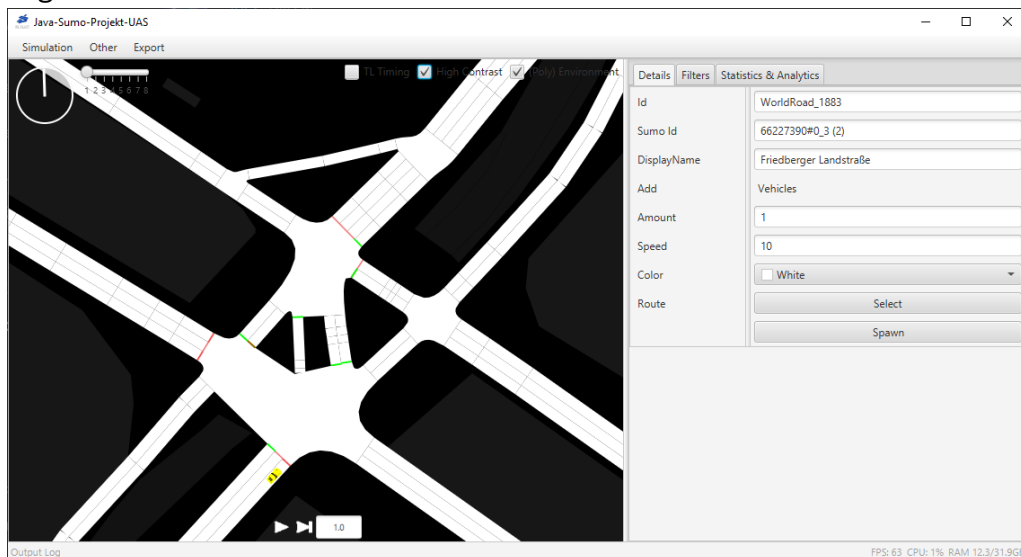
3.3 Miscellaneous controls

Checkboxes in the top right corner of the simulation window allow you to display the following options:

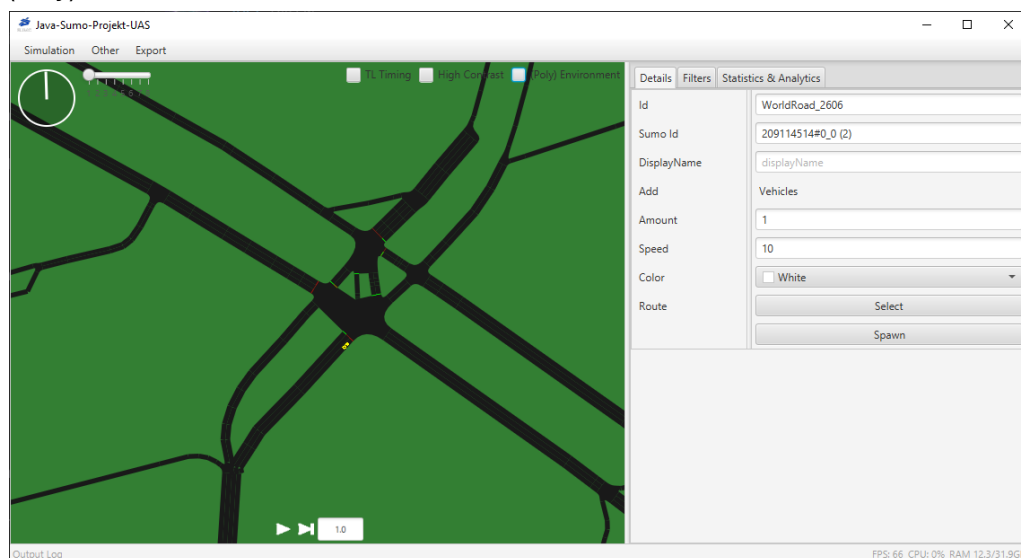
- TL Timing, which shows you when a traffic light will switch to another color next



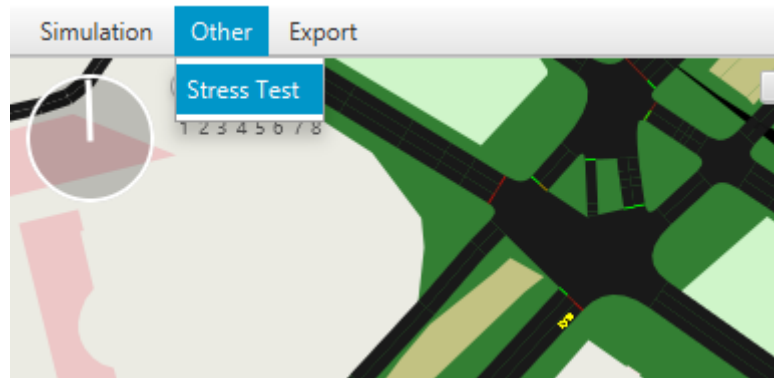
- High contrast mode



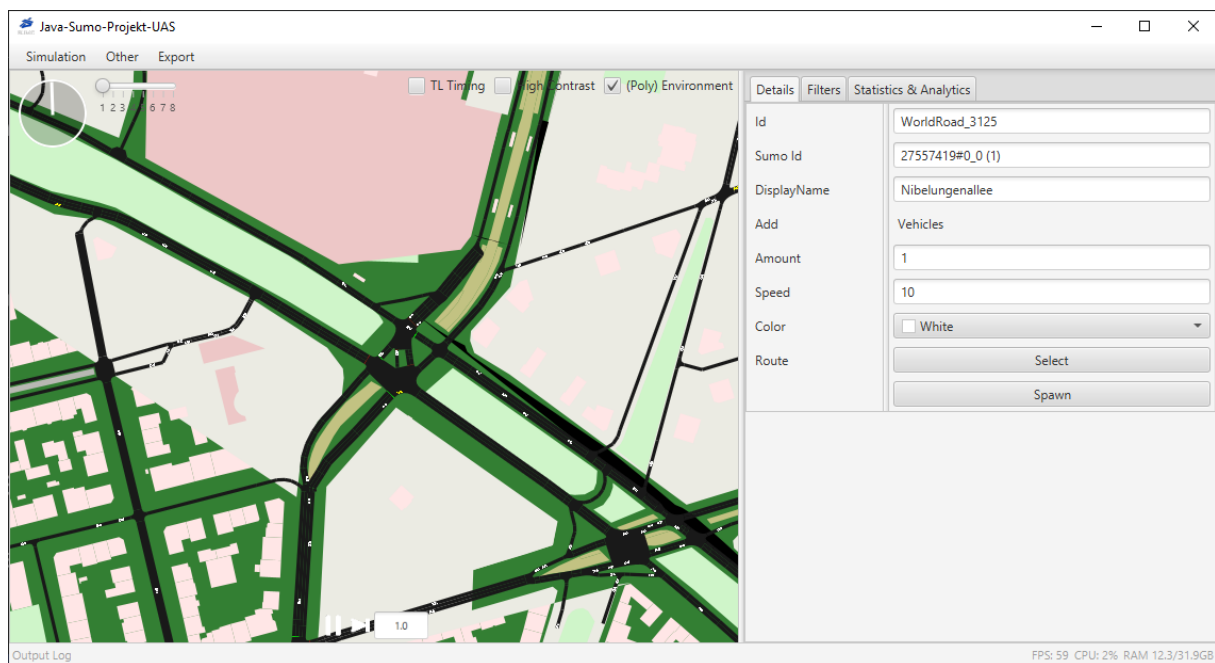
- (Poly) Environment



We also provide the user with a stress test feature which is located in the toolbar under the “Other” menu

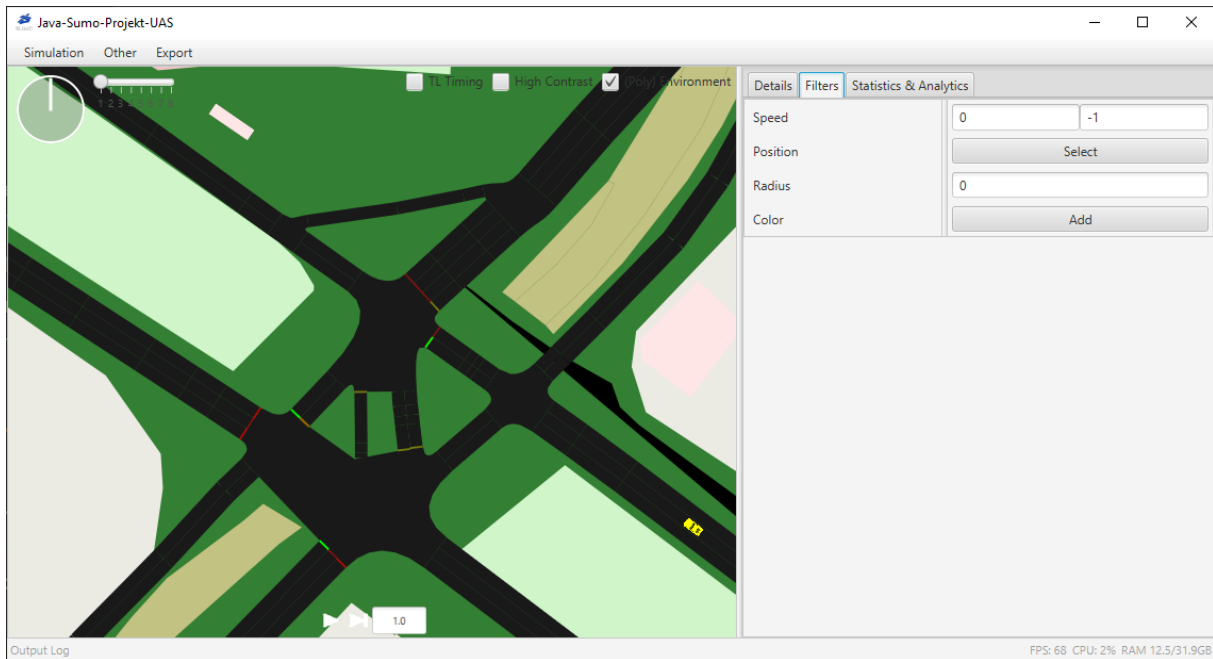


This spawns a large number of vehicles, dependent on the scale of the loaded network to test performance. Vehicles are spawned randomly.



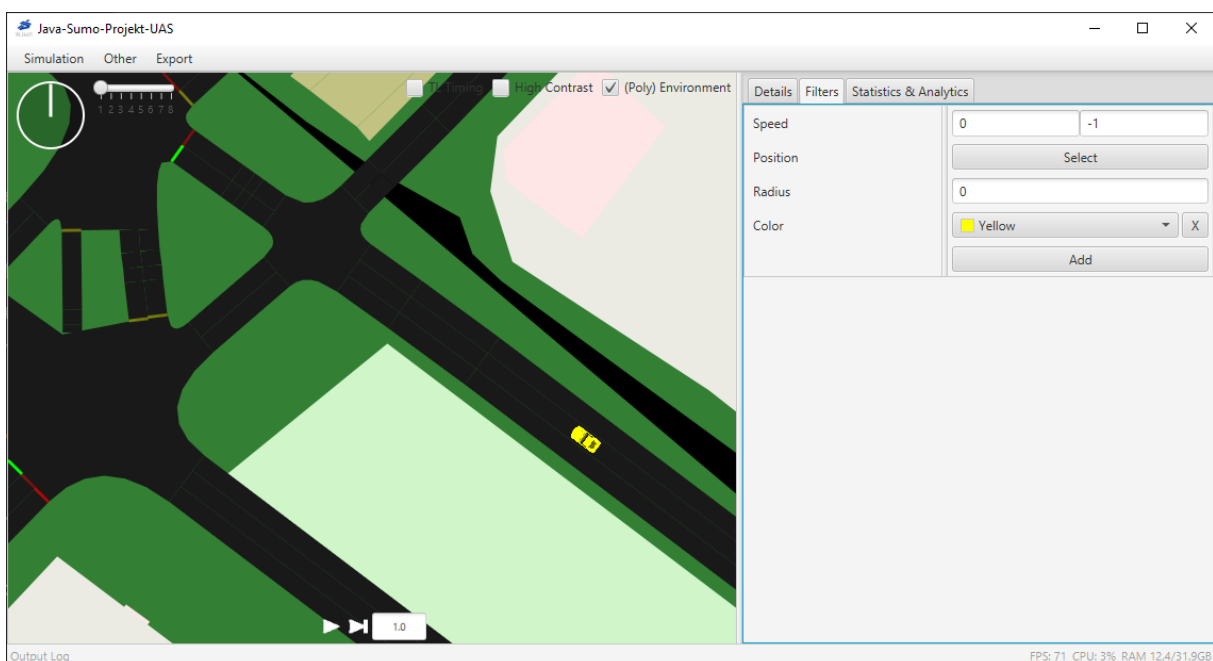
3.4 Filters

The filters tab allows you to, as the name implies, filter the vehicles for specific properties:



These include:

1. Speed → The left number is the slowest a car can go to be shown, and the right number is the maximum speed, -1 being unlimited, so all cars are shown by default
2. Position and Radius → Selecting a position and a radius will only display cars that are currently in that radius; 0 will display all cars on the map
3. Color → Only cars with the desired color will be displayed. By pressing the add button, you can add more colors to filter by. If you wish to remove the filter, use the “X” that is located next to the color



3.5 Statistics and Analytics

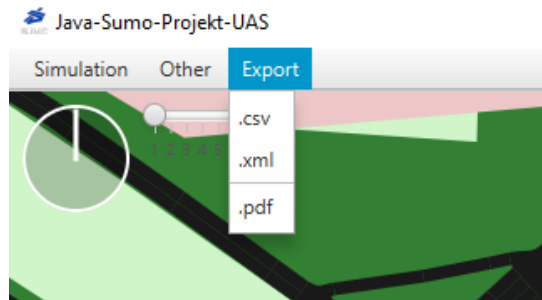
Under the “Statistics and Analytics” tab, you will find a graph, displaying the average speed of all vehicles in the simulation, as well as the occupancy ratio of the used streets with their length in meters. These metrics are updated every step, 0 being the current step and all other numbers being the steps before that in ascending order (10 means 10 steps before the current one).



4. Exporting

4.1 Export controls

If you have run your simulation and would now like to export your results, navigate to the “Export” menu in the toolbar.

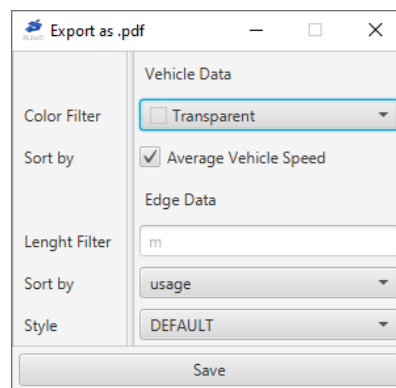


You will be prompted to choose with what format you want to export.

Available formats are:

- .csv
- .xml
- .pdf

Once you have selected your desired format, a window will appear, prompting you to choose some additional options:



Here you can choose:

Vehicle data:

1. Color filter → Only cars that have a specific color will be displayed in the export; leave transparent if all cars should be included
2. The option to sort these vehicles by average speed

Edge data:

1. Length Filter → Only roads that have a length of the number that was set or higher will be exported; leave blank for all roads
2. The option to sort by either usage (occupancy ratio), the name of the road or the length.

Lastly, you can choose from one of the provided styles that are: Default, Minimalistic, Young, Serious and no style. This option is only available for .pdf file exports.

Once all settings have been set to the desired parameters, click the “Save” button.

The program will create the file in a folder named “output” that is located in the same directory as the program. If no such folder is found, one will be created.